



Non-parametric identification and estimation of the number of components in multivariate mixtures

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Summary. We analyse the identifiability of the number of components in k -variate, M -component finite mixture models in which each component distribution has independent marginals, including models in latent class analysis. Without making parametric assumptions on the component distributions, we investigate how one can identify the number of components from the distribution function of the observed data. When $k \geq 2$, a lower bound on the number of components (M) is non-parametrically identifiable from the rank of a matrix constructed from the distribution function of the observed variables. Building on this identification condition, we develop a procedure to estimate a lower bound on the number of components consistently.

Keywords: Finite mixture; Latent class analysis; Non-negative rank; Rank estimation

1. Introduction

Finite mixture models provide flexible ways to model unobserved population heterogeneity. Because of their flexibility, finite mixtures have been used in numerous applications in diverse fields such as biological, physical and social sciences. Comprehensive theoretical accounts and examples of applications can be found in Everitt and Hand (1981), Titterton *et al.* (1985), McLachlan and Basford (1988), Lindsay (1995) and McLachlan and Peel (2000).

A finite mixture model is characterized by three main determinants: the number of components, the component distributions and the mixing proportions. As emphasized in Hettmansperger and Thomas (2000), there is often little theoretical guidance for selecting the number of components and/or the form of the component distributions despite their key role in the specification of mixtures. Furthermore, it has been known that the estimates of the number of components are sensitive to the choice of the component distributions (see, for example, Schork *et al.* (1990) and Roeder (1994)), and that imposing incorrect parametric restrictions on the component distributions may lead to erroneous inference on the number of components (Cruz-Medina *et al.*, 2004).

This paper analyses the non-parametric identifiability of the number of components in k -variate, M -component finite mixture models of $W = (W_1, \dots, W_k)'$ under the assumption that the W_j s are independently (but not necessarily identically) distributed within each component:

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$$F(w) = F(w_1, \dots, w_k) = \sum_{m=1}^{\tilde{M}} \pi^m F_1^m(w_1) F_2^m(w_2) \dots F_k^m(w_k), \quad \pi^m > 0, \quad \sum_{m=1}^{\tilde{M}} \pi^m = 1. \quad (1)$$

Here, $F(w)$ is the distribution function of W , π^m is the mixing proportion of the m th subpopulation and $F_j^m(w_j)$ is the distribution function of W_j conditional on being from the m th subpopulation. The number of components in $F(w)$, M , is defined as the smallest positive integer $\tilde{M}$ for which a finite mixture representation (1) can be found.

We analyse how one can recover the number of components M from the exact knowledge of the distribution function of observed variables $F(w_1, \dots, w_k)$ when no parametric assumptions are imposed on the component distributions. Non-parametric identifiability and estimation of finite mixtures has recently attracted increasing attention. Hall and Zhou (2003), Hall *et al.* (2005) and Allman *et al.* (2009) analysed non-parametric identifiability of component distributions and mixing proportions in model (1) under known M . (Kasahara and Shimotsu (2009) studied non-parametric identification of finite mixture dynamic discrete choice models that are widely used in econometrics.) In particular, Allman *et al.* (2009) showed that, if $k \geq 3$, model (1) is non-parametrically identifiable for any M if the $F_j^m(w_j)$ s are linearly independent. Hettmansperger and Thomas (2000) and Cruz-Medina *et al.* (2004) analysed the non-parametric identification and estimation of model (1) with independent and identically distributed marginals by partitioning the support of W_j into bins and transforming the data to multinomial vectors. Benaglia *et al.* (2009) and Levine *et al.* (2011) have developed algorithms for estimating model (1) non-parametrically by using kernels. However, no theoretical results on the identification of the number of components in model (1) are provided in the existing literature.

We show that a lower bound on the number of components M is identified without imposing any parametric assumptions if $k \geq 2$. Interestingly, this result holds despite the fact that the component distributions are not identifiable when $k = 2$ (see Clogg (1981) and Hall and Zhou (2003)). The lower bound is stated in terms of the rank of a matrix constructed from the (multinomial) distribution function of the observed data, where, for continuous variables, we transform each element of W to a discrete random variable by partitioning its support as in Elmore *et al.* (2004). We also illustrate the cases in which the bound is tight except possibly for a set of mixture models with zero Lebesgue measure, and, therefore, the bound is tight generically in the sense of Allman *et al.* (2009), page 3106. By estimating the rank of its empirical analogue, we develop a procedure to estimate a lower bound on the number of components consistently. Simulations illustrate that our procedure performs well.

The mixture model (1) assumes that the marginal distributions are independent conditional on belonging to a subpopulation. The conditional independence assumption may be viewed as a version of a standard repeated measures random-effects model, in which multivariate observations on an individual are often assumed to be independent conditional on the identity of the individual. Model (1) has important applications as demonstrated in some recent works on non-parametric mixture models as well as those on multinomial mixtures (e.g. Zhou *et al.* (2005), Dunson and Xing (2009) and Bhattacharya and Dunson (2012)) and encompasses models in latent class analysis that has been widely used in many fields including sociology, psychology and biostatistics (Lazarsfeld and Henry, 1968; Clogg, 1995; Hagenars and McCutcheon, 2002; Magidson and Vermunt, 2004; Skrondal and Rabe-Hesketh, 2004). Once an estimate of a lower bound of M has been obtained, one can use algorithms such as Benaglia *et al.* (2009) and Levine *et al.* (2011) to estimate the mixture model (1) non-parametrically, provided that the mixing proportions and the component distributions are identifiable.

Numerous methods to select the number of components have been proposed in a parametric setting (see, for example, Henna (1985), Leroux (1992), Lindsay and Roeder (1992), Windham

and Cutler (1992), Roeder (1994), Chen and Kalbfleisch (1996), Dacunha-Castelle and Gassiat (1999), Keribin (2000), James *et al.* (2001) and Woo and Sriram (2006)). Our proposed procedure requires the conditional independence assumption but makes no distributional assumptions on the components. Furthermore, our selection procedure is based on a statistic whose asymptotic distribution is χ^2 or can be easily simulated, and it does not require the estimation of a mixture model with a different number of components.

The remainder of the paper is organized as follows. Section 2 discusses the non-parametric identifiability of a lower bound on the number of components under $k \geq 2$. Section 3 introduces a procedure to test a lower bound on the number of mixture components. Section 4 reports simulation results, and empirical examples are provided in Section 5. The on-line supplementary appendix contains the proofs, mathematical details and detailed results from simulations and empirical examples.

The programs that were used to analyse the data can be obtained from

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2. Non-parametric identification of a lower bound on the number of components

2.1. Two-variable case

We first analyse the non-parametric identification of a *lower bound* on the number of components for the mixture model (1) with $k = 2$. For notational clarity, we use X and Y in place of W_1 and W_2 . Specifically, consider the following finite mixture models of variable (X, Y) :

$$F(x, y) = \sum_{m=1}^{\tilde{M}} \pi^m F_x^m(x) F_y^m(y), \quad \pi^m > 0, \quad \sum_{m=1}^{\tilde{M}} \pi^m = 1, \quad (2)$$

where $F_x^m(x)$ and $F_y^m(y)$ are the distribution functions of X and Y conditional on being from the m th subpopulation. No assumptions are imposed on $F_x^m(x)$ s and $F_y^m(y)$ s except that they are distribution functions. Define the number of components in $F(x, y)$, $\tilde{M}$, as the smallest positive integer $\tilde{M}$ for which a finite mixture representation (2) can be found.

We proceed to construct a partition Δ of the support of (X, Y) , and to form a matrix that represents the distribution of (X, Y) over Δ . Let $\mathcal{X}$ and $\mathcal{Y}$ denote the support of X and Y . Partition $\mathcal{X}$ and $\mathcal{Y}$ into $|\Delta_x|$ and $|\Delta_y|$ mutually exclusive and exhaustive subsets respectively, as $\Delta_x = \{\delta_1^x, \dots, \delta_{|\Delta_x|}^x\}$ and $\Delta_y = \{\delta_1^y, \dots, \delta_{|\Delta_y|}^y\}$, where $|S|$ denotes the number of elements in a set S . Define $\Delta = \Delta_x \times \Delta_y$. Given a partition Δ , collect the distributions of X and Y conditional on being from the m th subpopulation into a vector as

$$\begin{aligned} p_x^m &= (\Pr(X \in \delta_1^x | m), \dots, \Pr(X \in \delta_{|\Delta_x|}^x | m))', \\ p_y^m &= (\Pr(Y \in \delta_1^y | m), \dots, \Pr(Y \in \delta_{|\Delta_y|}^y | m))' \end{aligned} \quad (3)$$

respectively. The vectors p_x^m and p_y^m implicitly depend on Δ_x and Δ_y .

Arrange $\Pr(X \in \delta_a^x, Y \in \delta_b^y)$ for partition level $(a, b) = (1, 1), \dots, (|\Delta_x|, |\Delta_y|)$ into a $|\Delta_x| \times |\Delta_y|$ bivariate probability matrix as

$$P_\Delta = \begin{pmatrix} \Pr(X \in \delta_1^x, Y \in \delta_1^y) & \dots & \Pr(X \in \delta_1^x, Y \in \delta_{|\Delta_y|}^y) \\ \vdots & \ddots & \vdots \\ \Pr(X \in \delta_{|\Delta_x|}^x, Y \in \delta_1^y) & \dots & \Pr(X \in \delta_{|\Delta_x|}^x, Y \in \delta_{|\Delta_y|}^y) \end{pmatrix}. \quad (4)$$

Then, P_Δ represents the distribution of (X, Y) on the partition Δ and can be expressed in terms of π^m s, p_x^m s and p_y^m s as

$$P_{\Delta} = \sum_{m=1}^{\tilde{M}} \pi^m p_x^m p_y^{m'}, \quad \pi^m > 0, \quad \sum_{m=1}^{\tilde{M}} \pi^m = 1. \quad (5)$$

Equation (5) is a finite mixture model (2) that is restricted to the partition Δ .

For a partition Δ , define *the number of components in P_{Δ}* as the smallest integer $\tilde{M}$ such that the finite mixture representation (5) is possible. The number of components in P_{Δ} is closely related to the concept of *non-negative rank* that was developed by Cohen and Rothblum (1993). For a non-negative matrix A , its non-negative rank is denoted by $\text{rank}_+(A)$ and defined as the smallest number of non-negative rank 1 matrices such that A equals their sum. Because P_{Δ} is a non-negative matrix and the right-hand side of equation (5) is the sum of non-negative rank 1 matrices, by definition, the number of components in P_{Δ} is the non-negative rank of P_{Δ} .

The following proposition, which was originally from Cohen and Rothblum (1993), states the properties of the non-negative rank of P_{Δ} and its relation to the rank of P_{Δ} . Propositions 1(a), 1(b) and 1(c) correspond to lemma 2.3, theorem 4.1 and corollary 4.2 of Cohen and Rothblum (1993) respectively.

Proposition 1 (Cohen and Rothblum, 1993).

- (a) $\text{rank}(P_{\Delta}) \leq \text{rank}_+(P_{\Delta}) \leq \min\{|\Delta_x|, |\Delta_y|\}$.
- (b) If $\text{rank}(P_{\Delta}) \leq 2$, then $\text{rank}(P_{\Delta}) = \text{rank}_+(P_{\Delta})$.
- (c) If $|\Delta_x| \leq 3$ or $|\Delta_y| \leq 3$, then $\text{rank}_+(P_{\Delta}) = \text{rank}(P_{\Delta})$.

From proposition 1(a), $\text{rank}(P_{\Delta})$ gives a lower bound on the number of components in P_{Δ} whereas the number of support points of X and Y gives an upper bound on the number of identifiable components since $|\Delta_x| \leq |\mathcal{X}|$ and $|\Delta_y| \leq |\mathcal{Y}|$. It follows from proposition 1 that $\text{rank}_+(P_{\Delta}) = \text{rank}(P_{\Delta})$ if $\text{rank}_+(P_{\Delta}) \leq 3$.

The number of components in P_{Δ} is identified with the non-negative rank of P_{Δ} . Determining the non-negative rank of a matrix is computationally difficult however, and is still a subject of on-going research (see, for example, Dong *et al.* (2009)). (Vavasis (2009) has shown that determining the non-negative rank of a matrix is non-deterministic polynomial time hard.) Therefore, it is useful to characterize a *lower bound* on the number of components in P_{Δ} in terms of the rank of P_{Δ} .

An obvious limitation of the lower bound based on the rank of P_{Δ} is a possible discrepancy between the lower bound and the actual number of components. This is because the latter requires that the components π^m s, p_x^m s and p_y^m s in expression (5) are non-negative whereas the former does not. (For example, suppose that $|\Delta_x| = |\Delta_y| = 4$ and $P_{\Delta} = \sum_{m=1}^4 \pi^m p_x^m p_y^{m'}$, where $\pi^m > 0$ and p_x^m s are linearly independent but $p_y^1 + p_y^2 - p_y^3 - p_y^4 = 0$, so that $\text{rank}(P_{\Delta})$ is at most 3. Writing one p_y^m in terms of the other p_y^m s and substituting into P_{Δ} will give a three-term mixture representation of P_{Δ} . However, if $-\pi^1 p_x^1 + \pi^2 p_x^2$ and $-\pi^3 p_x^3 + \pi^4 p_x^4$ have both positive and negative elements, then the resulting three-term mixture representation necessarily contains negative components, and $\text{rank}_+(P_{\Delta})$ is strictly larger than 3.) We investigate the size of a set of mixture models wherein $\text{rank}_+(P_{\Delta}) > \text{rank}(P_{\Delta})$. Given a positive integer M_0 , define the space of M_0 -component mixture models $\theta = \{p_x^m, p_y^m, \pi^m\}_{m=1}^{M_0}$ by $\Theta \subset \mathcal{S}_{|\Delta_x|-1}^{M_0} \times \mathcal{S}_{|\Delta_y|-1}^{M_0} \times \mathcal{S}_{M_0-1}$ as in Allman *et al.* (2009), page 3107, where $\mathcal{S}_k$ denotes the standard k -simplex. The following proposition shows that if we randomly draw a mixture model θ from Θ and construct a matrix $P(\theta) = \sum_{m=1}^{M_0} \pi^m p_x^m p_y^{m'}$, then we have $\text{rank}\{P(\theta)\} = M_0$ with probability 1. This result holds because, when $\text{rank}\{P(\theta)\} < M_0$, either the vectors $\{p_x^m\}_{m=1}^{M_0}$ or $\{p_y^m\}_{m=1}^{M_0}$ are linearly dependent, but the set of linearly dependent $\{p_x^m\}_{m=1}^{M_0}$ s (or $\{p_y^m\}_{m=1}^{M_0}$ s) has Lebesgue measure 0 in $\mathbb{R}^{\Delta_x \times M_0}$ (or $\mathbb{R}^{\Delta_y \times M_0}$).

Proposition 2. If $M_0 \leq \min\{|\Delta_x|, |\Delta_y|\}$, then $M_0 = \text{rank}_+\{P(\theta)\} = \text{rank}\{P(\theta)\}$ holds for all the points in Θ except possibly for a set of Lebesgue measure 0.

When X and Y are discrete, taking the support of (X, Y) as Δ and applying proposition 2 gives that $\text{rank}(P_\Delta) = \text{rank}_+(P_\Delta) = M$ with probability 1 if we draw an M -component bivariate mixture model with conditionally independent marginals. Hence, the bound is tight with probability 1. When X and Y are continuous, there is no obvious choice of a single partition Δ . The non-negative rank of P_Δ could be strictly smaller than M when a single partition Δ does not fully reveal the information for identifying the number of components in $F(x, y)$. A tighter lower bound of M may be obtained by taking the maximum value of the rank of P_Δ s across different partitions.

Proposition 3. Suppose that, in model (2), the distribution of (X, Y) is continuous.

- (a) If $\{F_x^m(\cdot)\}_{m=1}^M$ are linearly independent and $\{F_y^m(\cdot)\}_{m=1}^M$ are linearly independent, then there exists a partition Δ such that $\text{rank}(P_\Delta) = M$.
- (b) For any fixed Δ , the probability that $\text{rank}_+(P_\Delta) = \text{rank}(P_\Delta)$ is 1 when a mixture model is drawn randomly.

Proposition 3(a) gives a sufficient condition under which the rank of P_Δ is equal to M for some choice of Δ ; in this case, M is identified with the maximum value of $\text{rank}(P_\Delta)$ s over all possible partitions of $\mathcal{X} \times \mathcal{Y}$. Proposition 3(b) implies that whether $\text{rank}(P_\Delta) = M$ holds depends on whether $\text{rank}_+(P_\Delta) = M$ holds.

2.2. General k -variable case

We now illustrate how our approach in Section 2.1 can be applied to the mixture model (1) with $k \geq 3$ to obtain a lower bound on M . Consider a hyperrectangle partition $\Delta = \Delta_1 \times \dots \times \Delta_k$ of $\mathcal{W}$. Let P_Δ denote $F(w)$ on Δ . Note that P_Δ is a k -dimensional array. P_Δ is written as a weighted average of k -dimensional tensors as follows:

$$P_\Delta = \sum_{m=1}^{\text{rank}_+(P_\Delta)} \pi^m P^m, \quad P^m = \otimes_{j=1}^k p_j^m, \quad \pi^m > 0, \quad \sum_{m=1}^{\text{rank}_+(P_\Delta)} \pi^m = 1, \quad (6)$$

where ' $\otimes$ ' denotes a tensor product and p_j^m is a $|\Delta_j| \times 1$ vector. Here, $\text{rank}_+(P_\Delta)$ is defined as the smallest positive integer for which a representation (6) holds and is called the non-negative (tensor) rank of P_Δ (see, for example, Lim and Common (2009)). As in the two-variable case, $\text{rank}_+(P_\Delta)$ provides a lower bound on M .

We construct a matrix from P_Δ by grouping the variables in $W = (W_1, \dots, W_k)'$ into two groups. We index the groupings by α . For the grouping α , let X^α and Y^α be the grouped variables. Let $\Delta_{x^\alpha} = \{\delta_1^{x^\alpha}, \dots, \delta_{|\Delta_{x^\alpha}|}^{x^\alpha}\} = \Pi_{j \in S_x(\alpha)} \Delta_j$ be the partition of the support of X^α , where $S_x(\alpha)$ is the set of indices such that $W_j \in X^\alpha$, and define Δ_{y^α} similarly. Then, we construct a $|\Delta_{x^\alpha}| \times |\Delta_{y^\alpha}|$ bivariate probability matrix P_Δ^α by arranging $\Pr(X^\alpha \in \delta_a^{x^\alpha}, Y^\alpha \in \delta_b^{y^\alpha})$ for partition level $(a, b) = (1, 1), \dots, (|\Delta_{x^\alpha}|, |\Delta_{y^\alpha}|)$ as in equation (4).

A lower bound on M can be obtained in terms of $\text{rank}_+(P_\Delta)$, $\text{rank}_+(P_\Delta^\alpha)$ and $\text{rank}(P_\Delta^\alpha)$ as $M \geq \text{rank}_+(P_\Delta) \geq \text{rank}_+(P_\Delta^\alpha) \geq \text{rank}(P_\Delta^\alpha)$. Taking the maximum value of $\text{rank}_+(P_\Delta)$, $\text{rank}_+(P_\Delta^\alpha)$ and $\text{rank}(P_\Delta^\alpha)$ across different partitions Δ s and different groupings α s gives tighter lower bounds. Such bounds may still be, however, strictly smaller than M .

We investigate when $\text{rank}_+(P_\Delta) = \text{rank}(P_\Delta^\alpha)$ holds. With a slight abuse of notation, given a positive integer M_0 , define the space of M_0 -component mixture models $\theta = \{p_1^m, \dots, p_k^m, \pi^m\}_{m=1}^{M_0}$ as $\Theta \subset \mathcal{S}_{|\Delta_1|-1}^{M_0} \times \dots \times \mathcal{S}_{|\Delta_k|-1}^{M_0} \times \mathcal{S}_{M_0-1}$. Given an element θ of Θ , group W into X^α and Y^α and

construct a bivariate probability matrix $P^\alpha(\theta) = \sum_{m=1}^{M_0} \pi^m p_{x^\alpha}^m p_{y^\alpha}^{m'}$, where $p_{x^\alpha}^m$ is $|\Delta_{x^\alpha}| \times 1$ and $p_{y^\alpha}^m$ is $|\Delta_{y^\alpha}| \times 1$. The following proposition shows that, if the grouped variables have sufficiently large state spaces relative to M_0 , a result analogous to proposition 2 holds for a k -variable model.

Proposition 4. If $M_0 \leq \min\{|\Delta_{x^\alpha}|, |\Delta_{y^\alpha}|\}$, then $M_0 = \text{rank}\{P^\alpha(\theta)\}$ holds for all the points in Θ except possibly for a set of Lebesgue measure 0.

In the continuous variable case, we have a simple corollary of proposition 3(a).

Corollary 1. Suppose that, in model (1), the distribution of W is continuous. If there is a grouping α such that the two families $\{F_{x^\alpha}^m(\cdot)\}_{m=1}^M$ and $\{F_{y^\alpha}^m(\cdot)\}_{m=1}^M$ are respectively composed of linearly independent vectors, then $\text{rank}(P_\Delta^\alpha) = M$.

2.3. Relation to latent class analysis

Consider a special case in which an observation vector $W = (W_1, \dots, W_k)'$ consists of k dichotomous or polytomous responses, which are typically answers to questions or results of diagnoses. In this case, our model (1) becomes identical to the model that is used in *latent class analysis*. For recent surveys and applications of latent class analysis, see the references in Section 1.

The latent class analysis with $k = 2$ (two-way contingency table) is also known as *latent budget analysis* (Goodman, 1974; Clogg, 1981; de Leeuw and van der Heijden, 1988). Testing the number of components in a latent budget model is particularly difficult because the parameters of the model are not identified unless some restrictions are imposed. Using our result, it is possible to identify a lower bound on M without imposing restrictions on the parameters, even though identifying a lower bound of M does not solve the problem of parameter non-identification.

3. Estimating a lower bound on the number of components

Proposition 1 in Section 2 shows that the rank of a matrix P_Δ in expression (5) gives a lower bound on the number of mixture components. In this section, we develop procedures to estimate the rank of P_Δ for a given partition Δ and extend these procedures to the case when there are more than two variables.

3.1. Statistic of Kleibergen and Paap (2006)

Kleibergen and Paap (2006) developed a procedure to test the null hypothesis that the rank of P_Δ is equal to r as described below. For notational brevity, let $s = |\Delta_x|$ and $t = |\Delta_y|$, and write the singular value decomposition of an $s \times t$ matrix P_Δ as

$$P_\Delta = USV' = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix} \begin{pmatrix} S_1 & 0 \\ 0 & S_2 \end{pmatrix} \begin{pmatrix} V'_{11} & V'_{12} \\ V'_{21} & V'_{22} \end{pmatrix},$$

where U is a $t \times t$ orthogonal matrix, V is a $t \times t$ orthogonal matrix and S is an $s \times t$ matrix that contains the singular values of P_Δ in decreasing order on its main diagonal and is equal to 0 elsewhere. In the partition of U , S and V on the right-hand side, U_{11} , S_1 and V_{11} are $r \times r$, and the dimensions of the other submatrices are defined conformably. Then, the null hypothesis $H_0: \text{rank}(P_\Delta) = r$ is equivalent to $H_0: S_2 = 0$ because the rank of a matrix is equal to the number of non-zero singular values.

The statistic of Kleibergen and Paap (2006) is based on an orthogonal transformation of S_2 given by $\Lambda_r = A'_{r,\perp} P_\Delta B'_{r,\perp}$, where

$$A'_{r,\perp} = (U_{22}U'_{22})^{1/2}(U'_{22})^{-1}(U'_{12}|U'_{22})$$

and

$$B_{r,\perp} = (V_{22}V'_{22})^{1/2}(V'_{22})^{-1}(V'_{12}|V'_{22}).$$

Unlike S_2 , Λ_r is not restricted to be non-negative. Then, the null hypothesis $H_0 : \text{rank}(P_\Delta) = r$ is equivalent to $H_0 : \Lambda_r = 0$. Let $\hat{P}_\Delta$ be an estimator of the matrix P_Δ with sample size N . We assume that $\text{vec}(\hat{P}_\Delta)$ is asymptotically normally distributed.

Assumption 1. $N^{1/2} \text{vec}(\hat{P}_\Delta - P_\Delta) \rightarrow_d N(0, \Sigma)$ as $N \rightarrow \infty$, where Σ is an $st \times st$ covariance matrix.

When the distribution of W is discrete or Δ is predetermined, $\text{vec}(\hat{P}_\Delta)$ follows a multinomial distribution, and a formula for Σ is easily available. If W has a continuous distribution and the empirical quantiles of the W_j s are used to construct Δ , then $\text{vec}(\hat{P}_\Delta)$ follows the empirical multivariate quantile-partitioned distribution (Borkowf, 2000) that is described in the on-line supplemental appendix, and one can use the bootstrap to estimate Σ .

We estimate Λ_r by $\hat{\Lambda}_r = \hat{A}'_{r,\perp} \hat{P}_\Delta \hat{B}'_{r,\perp}$ and test $H_0 : \Lambda_r = 0$, where $\hat{A}_{r,\perp}$ and $\hat{B}_{r,\perp}$ are the estimators of $A_{r,\perp}$ and $B_{r,\perp}$ obtained from the singular value decomposition of $\hat{P}_\Delta$. Kleibergen and Paap (2006) derived the asymptotic distribution of $\hat{\lambda}_r = \text{vec}(\hat{\Lambda}_r)$, as summarized below.

Proposition 5 (Kleibergen and Paap (2006), theorem 1). Suppose that assumption 1 holds and that $\Omega_r = (B_{r,\perp} \otimes A'_{r,\perp}) \Sigma (B_{r,\perp} \otimes A'_{r,\perp})'$ is non-singular. If $\text{rank}(P_\Delta) \leq r$, then $N^{1/2} \hat{\lambda}_r \rightarrow_d N(0, \Omega_r)$ as $N \rightarrow \infty$.

Kleibergen and Paap (2006), corollary 1, proposed the statistic called the *rk-statistic*:

$$\text{rk}(r) = N \hat{\lambda}'_r \hat{\Omega}_r^{-1} \hat{\lambda}_r, \quad (7)$$

where $\hat{\Omega}_r$ is a consistent estimator for Ω_r . If the assumptions of proposition 5 hold, then $\text{rk}(r)$ converges in distribution to a $\chi^2\{(s-r)(t-r)\}$ random variable under $H_0 : \text{rank}(P_\Delta) = r$. The non-singularity assumption on Ω_r can be relaxed by using the Moore–Penrose pseudoinverse as discussed in Section 3.4.

The choice of Δ is left to the researcher. As for the number of partitions, it is desirable to use a partition that is as fine as possible from the perspective of pure identification, but using a finer partition increases the variance of $\hat{P}_\Delta$. In practice, we suggest setting the number of partitions equal to 1 plus the maximum number of components that we want to allow for in modelling the data. As for the choice of partitions, a natural choice would be to use equiprobable intervals as in Pearson's χ^2 -test, but there may be cases where using a non-equiprobable partition gives a stronger power because mixture models often have fat tails. The optimal choice of partitions remains an open question.

3.2. Sequential hypothesis testing

Denote the population rank of P_Δ by r_0 . To estimate r_0 , we sequentially test $H_0 : \text{rank}(P_\Delta) = r$ against $H_1 : \text{rank}(P_\Delta) > r$ starting from $r = 0$, and then $r = 1, \dots, t^*$, where $t^* = \min\{s, t\}$. The first value for r that leads to a non-rejection of H_0 gives our estimate for r_0 .

For $r = 0, \dots, t^*$, let $c'_{1-\alpha_N}$ denote the $100(1 - \alpha_N)$ percentile of the cumulative distribution function of a $\chi^2\{(s-r)(t-r)\}$ random variable. Then, our estimator based on the sequential hypothesis test (SHT) is defined as

$$\hat{r} = \min_{r \in \{0, \dots, t^*\}} \{r : \text{rk}(i) \geq c_{1-\alpha_N}^i, i=0, \dots, r-1, \text{rk}(r) < c_{1-\alpha_N}^r\}. \quad (8)$$

The estimator $\hat{r}$ depends on the choice of the significance level α_N . As shown by Robin and Smith (2000), theorem 5.2, $\hat{r}$ converges to r_0 in probability as $N \rightarrow \infty$ if we choose α_N such that $\alpha_N = o(1)$ and $-N^{-1} \log(\alpha_N) = o(1)$.

3.3. Information criteria

We also consider a selection procedure by information criteria to estimate r_0 consistently. Consider the criterion function $Q(r) = \text{rk}(r) - f(N)g(r)$, where $g(r)$ is a (possibly stochastic) penalty function. Define $\tilde{r} = \arg \min_{1 \leq r \leq t^*} Q(r)$. Under a standard condition on $f(N)$ and $g(r)$, this gives a consistent estimate of r_0 .

Proposition 6. Suppose that the conditions of proposition 5 hold, and $\hat{\Omega}_r$ converges to a non-singular matrix for any $r \geq r_0$. Suppose that $f(N) \rightarrow \infty$, $f(N)/N \rightarrow 0$ and $\Pr\{g(r) - g(r_0) < 0\} \rightarrow 1$ for all $r > r_0$ as $N \rightarrow \infty$. Then, $\tilde{r} \rightarrow_p r_0$.

For the choice of $f(N)$ and $g(r)$, we consider the penalty terms in the Akaike information criterion (AIC), the Bayesian information criterion (BIC) and the Hannan–Quinn (HQ) information criterion. We choose $g(r) = (s-r)(t-r)$ with $f(N) = 2$ for the AIC, $f(N) = \log(N)$ for the BIC and $f(N) = 2 \log\{\log(N)\}$ for the HQ criterion. The BIC and HQ model selection procedures provide a consistent estimate of r_0 since their choice of $f(N)$ and $g(r)$ satisfies the conditions in proposition 6. In contrast, the AIC is not necessarily consistent and tends to overestimate r_0 with a large sample size.

3.4. Case of multiple variables

Suppose that $W = (W_1, \dots, W_k)'$ with $k \geq 3$ follows the distribution function (1). As in Section 2.2, we group the variables in W into two groups X^α and Y^α , with the grouping index α , and let P_Δ^α denote a $|\Delta_{X^\alpha}| \times |\Delta_{Y^\alpha}|$ bivariate probability matrix derived from the joint distribution of X^α and Y^α on a partition Δ . We test the null hypothesis that $\text{rank}(P_\Delta^\alpha) \leq r$ for all $\alpha \in \mathcal{A}_0$, where $\mathcal{A}_0$ is a set of the α s over which we construct test statistics.

We assume that all the variables in W are included in the first grouping $\{X^1, Y^1\}$. Then, for every $\alpha \in \mathcal{A}_0$, the elements of the probability matrix P_Δ^α can be expressed as a linear combination of the elements of P_Δ^1 and, therefore, there is a matrix Π^α such that $\text{vec}(P_\Delta^\alpha) = \Pi^\alpha \text{vec}(P_\Delta^1)$.

Define $A_{r,\perp}^\alpha$, $B_{r,\perp}^\alpha$ and λ_r^α analogously to $A_{r,\perp}$, $B_{r,\perp}$ and λ_r in Section 4.1 by using P_Δ^α in place of P_Δ . Define $\hat{\lambda}_r^\alpha = \text{vec}\{(\hat{A}_{r,\perp}^\alpha)' \hat{P}_\Delta^\alpha (\hat{B}_{r,\perp}^\alpha)'\} = (\hat{B}_{r,\perp}^\alpha \otimes (\hat{A}_{r,\perp}^\alpha)') \Pi^\alpha \text{vec}(\hat{P}_\Delta^1)$ by using the estimators of P_Δ^1 , $A_{r,\perp}^\alpha$ and $B_{r,\perp}^\alpha$. To test the null hypothesis that $\text{rank}(P_\Delta^\alpha) \leq r$ for all $\alpha \in \mathcal{A}_0$, we stack $\hat{\lambda}_r^\alpha$ s into a vector as $\hat{\lambda}_r(\mathcal{A}_0) = ((\hat{\lambda}_r^1)', \dots, (\hat{\lambda}_r^{|\mathcal{A}_0|})')'$ and test the null hypothesis $\lambda_r(\mathcal{A}_0) = 0$. Extending proposition 5, the following corollary establishes the asymptotic normality of $\hat{\lambda}_r(\mathcal{A}_0)$. We omit its proof for brevity, because it is a straightforward consequence of Slutsky's theorem.

Corollary 2. Suppose that $N^{1/2} \text{vec}(\hat{P}_\Delta^1 - P_\Delta^1) \rightarrow_d N(0, \Sigma_\Delta^1)$ and that $\Omega_r(\mathcal{A}_0)$ defined in equation (9) below is non-singular. If $\text{rank}(P_\Delta^\alpha) \leq r$ for all $\alpha \in \mathcal{A}_0$, we have $N^{1/2} \hat{\lambda}_r(\mathcal{A}_0) \rightarrow_d N\{0, \Omega_r(\mathcal{A}_0)\}$ as $N \rightarrow \infty$, where

$$\Omega_r(\mathcal{A}_0) = \begin{pmatrix} \Psi^1 \Sigma_\Delta^1 (\Psi^1)' & \dots & \Psi^1 \Sigma_\Delta^1 (\Psi^{|\mathcal{A}_0|})' \\ \vdots & \ddots & \vdots \\ \Psi^{|\mathcal{A}_0|} \Sigma_\Delta^1 (\Psi^1)' & \dots & \Psi^{|\mathcal{A}_0|} \Sigma_\Delta^1 (\Psi^{|\mathcal{A}_0|})' \end{pmatrix} \quad (9)$$

and $\Psi^\alpha = (B_{r,\perp}^\alpha \otimes (A_{r,\perp}^\alpha)') \Pi^\alpha$.

We can test the null hypothesis $H_0 : \text{rank}(P_\Delta^\alpha) \leq r$ for all $\alpha \in \mathcal{A}_0$ by the *average rk-statistic* defined as

$$\text{ave-rk}(r, \mathcal{A}_0) = N(\hat{\lambda}_r(\mathcal{A}_0))' \hat{\Omega}_r(\mathcal{A}_0)^{-1} \hat{\lambda}_r(\mathcal{A}_0), \quad (10)$$

where $\hat{\Omega}_r(\mathcal{A}_0)$ is a consistent estimator of $\Omega_r(\mathcal{A}_0)$. Thus, $\text{ave-rk}(r, \mathcal{A}_0)$ combines information from $\hat{\lambda}_r^\alpha$'s across different α s by using the inverse of their covariance matrix as the weight. Under the assumptions in corollary 2, $\text{ave-rk}(r, \mathcal{A}_0)$ converges in distribution to a $\chi^2\{\nu(\mathcal{A}_0)\}$ random variable, where $\nu(\mathcal{A}_0) \equiv \sum_{\alpha \in \mathcal{A}_0} (|\Delta_{x^\alpha}| - r)(|\Delta_{y^\alpha}| - r)$ is the number of elements in $\hat{\lambda}_r(\mathcal{A}_0)$. We note, however, that the average rk-statistic may give a slack lower bound when enumerating sufficiently many of the groupings and partitions of the data are not computationally feasible.

When $\nu(\mathcal{A}_0)$ is larger than the rank of Σ_Δ^1 , the covariance matrix $\Omega_r(\mathcal{A}_0)$ becomes singular and the assumption of corollary 2 is violated. In such a case, if $\Pr[\text{rank}\{\hat{\Omega}_r(\mathcal{A}_0)\} = \text{rank}[\Omega_r(\mathcal{A}_0)] \rightarrow 1$, using the Moore–Penrose pseudoinverse of $\hat{\Omega}_r(\mathcal{A}_0)$ in the ave-rk -statistic (10) gives a test statistic whose asymptotic distribution is $\chi^2[\text{rank}\{\Omega_r(\mathcal{A}_0)\}]$ (Andrews, 1987). However, in finite samples, if $\hat{\Omega}_r(\mathcal{A}_0)$ has a very small but non-zero eigenvalue, its pseudoinverse may take a very large value and behave erratically. To deal with the singularity of $\Omega_r(\mathcal{A}_0)$, we follow Lütkepohl and Burda (1997) to use a suitable reduced rank estimator in place of $\hat{\Omega}_r(\mathcal{A}_0)$. Given a small constant c , we apply a singular decomposition to $\hat{\Omega}_r(\mathcal{A}_0)$ and replace the eigenvalues that are smaller than c with 0. Let $\hat{\Omega}_{r,c}(\mathcal{A}_0)$ denote this low rank approximation of $\hat{\Omega}_r(\mathcal{A}_0)$, and define the *modified average rk-statistic* as

$$\text{ave-rk}^+(r, \mathcal{A}_0) = N(\hat{\lambda}_r(\mathcal{A}_0))' (\hat{\Omega}_{r,c}(\mathcal{A}_0))^+ \hat{\lambda}_r(\mathcal{A}_0). \quad (11)$$

The asymptotic distribution of $\text{ave-rk}^+(r, \mathcal{A}_0)$ is $\chi^2(J_c)$, where J_c is the number of eigenvalues of $\Omega_r(\mathcal{A}_0)$ that are no smaller than c . The behaviour of $\text{ave-rk}^+(r, \mathcal{A}_0)$ could be sensitive to the choice of c . In the simulations in Section 5, we set c equal to 0.01 times the largest eigenvalue of $\Omega_r(\mathcal{A}_0)$. (See Lütkepohl and Burda (1997) for other choices of c .)

We also consider an alternative statistic that is applicable even when $\nu(\mathcal{A}_0)$ is large. In the alternative statistic, we first choose K subsets of $\mathcal{A}_0$ as $\{A_1, \dots, A_K\}$ so that $\mathcal{A}_0 = \bigcup_{j=1}^K A_j$, and construct the $\text{ave-rk}^+(r, A_j)$ as in equation (11) but using A_j in place of $\mathcal{A}_0$. We then combine the information in $\text{ave-rk}^+(r, A_j)$ for $j = 1, \dots, K$ into the modified max-rk-statistic defined as $\text{max-rk}^+(r) = \max_{j=1, \dots, K} \text{ave-rk}^+(r, A_j)$. By choosing A_j s so that the degree of freedom $\nu(A_j)$ is sufficiently small, $\text{max-rk}^+(r)$ would be less sensitive to the choice of c than $\text{ave-rk}^+(r, \mathcal{A}_0)$. We can apply the sequential hypothesis testing procedure to $\text{max-rk}^+(r)$. Its asymptotic null distribution is not χ^2 but it can be easily simulated by using the relationship $N^{1/2} \hat{\lambda}_r^\alpha = N^{1/2} \hat{\Psi}^\alpha \{\text{vec}(\hat{P}_\Delta^1 - P_\Delta^1)\}$.

4. Simulation study

We conduct simulation experiments to assess the finite sample performance of our proposed procedures for selecting the number of components. The reported results are based on 1000 simulated samples from mixtures with $M = 3$ components with three different sample sizes: $N = 500, 2000, 8000$. To construct the rk-statistic (7) and the ave-rk^+ -statistic (11) for each sample, we estimate Ω_r and $\Omega_r(\mathcal{A}_0)$ consistently by the non-parametric bootstrap using 1000 random samples with replacement from empirical distributions.

In the first experiment, we generate samples of (X, Y) from a three-component normal mixture $\sum_{m=1}^3 \pi^m N_2(\mu^m, I_2)$ with $\mu^1 = (0, 0)'$, $\mu^2 = (1.0, 2.0)'$, $\mu^3 = (2.0, 1.0)'$ and $\pi^1 = \pi^2 = \pi^3 = \frac{1}{3}$. We denote the 100q-percentiles of empirical distributions of X and Y by z_q^x and z_q^y respectively. We con-

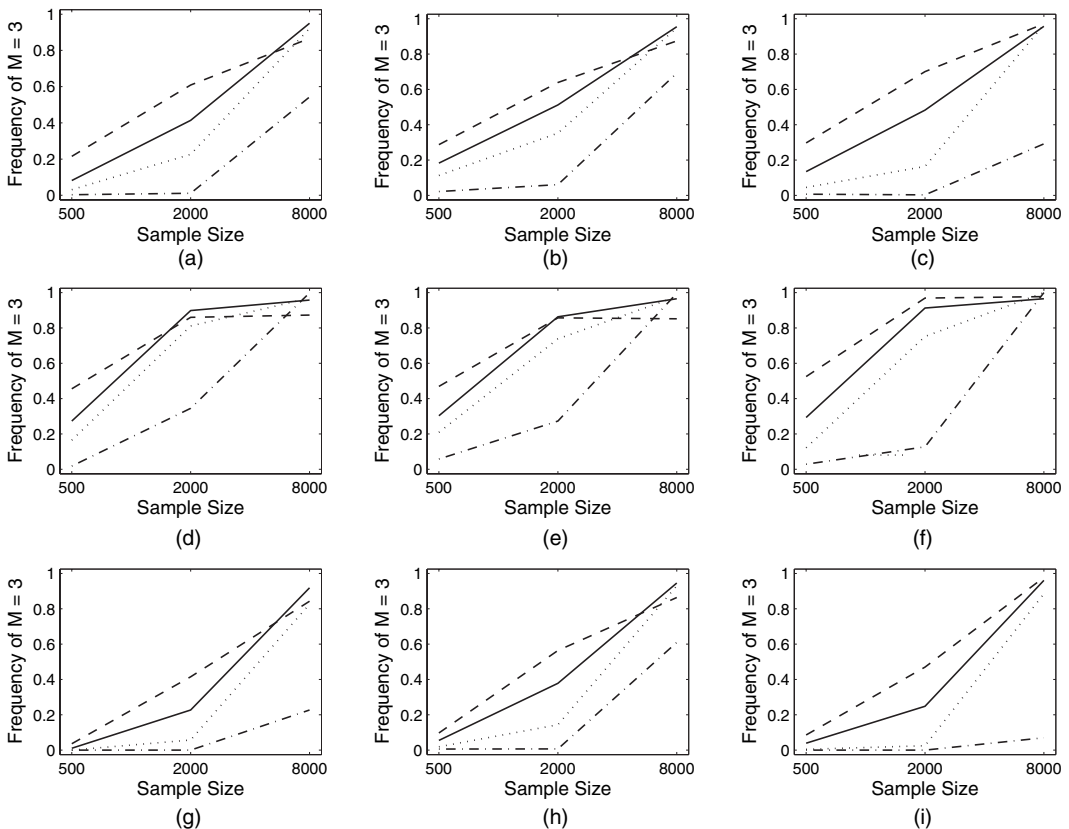


Fig. 1. Selection frequencies of the number of components—two variables (—, SHT, $\alpha = 0.05$; ---, AIC; - · - · -, BIC; · · · · ·, HQ): (a) normal, partition 1; (b) normal, partition 2; (c) normal, partition 3; (d) χ^2 , partition 1; (e) χ^2 , partition 2; (f) χ^2 , partition 3; (g) gamma, partition 1; (h) gamma, partition 2; (i) gamma, partition 3

sider three different partitions of the form $\Delta^j = \{(-\infty, z_{q_1}^j], (z_{q_1}^j, z_{q_2}^j], \dots, (z_{q_{t-2}}^j, z_{q_{t-1}}^j], (z_{q_{t-1}}^j, \infty)\}$ for $j = x, y$, where t is the number of partitions. We choose $t = 4$ with $(q_1, q_2, q_3) = (0.25, 0.5, 0.75)$ for partition 1 and $(q_1, q_2, q_3) = (0.1, 0.5, 0.9)$ for partition 2. Thus, the support of X or Y is partitioned into four (asymptotically) equiprobable subsets in partition 1 whereas partition 2 provides finer partitions in the tail part of distributions than partition 1. Partition 3 combines partitions 1 and 2 as $(q_1, q_2, q_3, q_4, q_5) = (0.1, 0.25, 0.5, 0.75, 0.9)$.

Figs 1(a), 1(b) and 1(c) show the frequency that the SHT with $\alpha = 0.05$, AIC, BIC and HQ based on the rk-statistic correctly select $M = 3$ for partitions 1, 2 and 3 respectively. Across different partitions, the performance of all the procedures improves as the sample size increases. With sample sizes 500 and 2000, the AIC outperforms the other statistics but overestimates the number of components at $N = 8000$ in partitions 1 and 2. The BIC exhibits the worst performance among all of the methods in this set-up. The HQ criterion is a better choice than the BIC but is outperformed by the SHT in most cases. The SHT in partition 2 performs better than in partition 1, even though the improvement is not substantial. This is probably because the tail part of distributions provides important information for separately identifying different components in this experiment. The SHT in partition 2 also performs better than the SHT

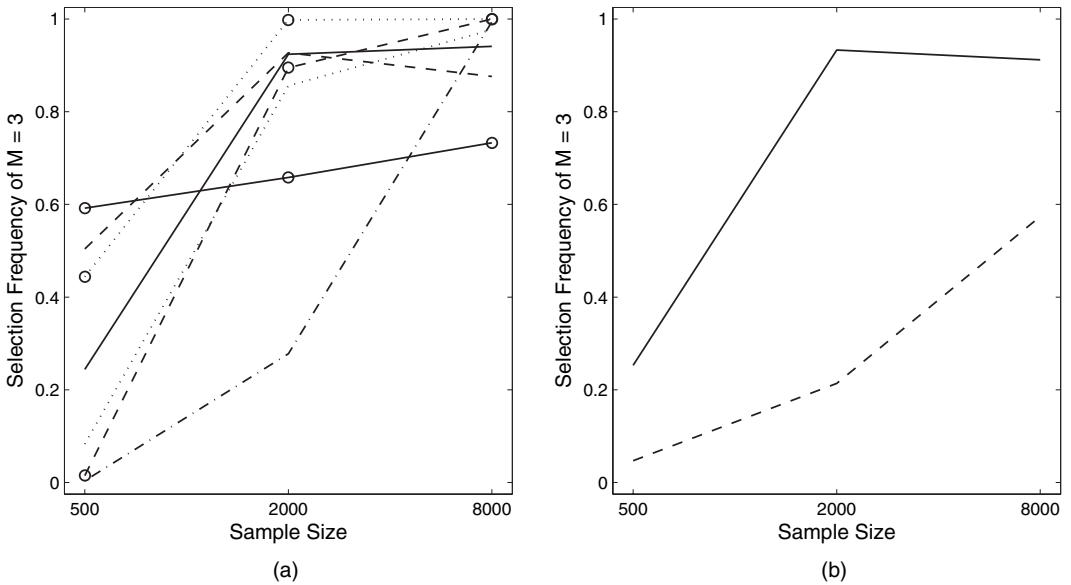


Fig. 2. Selection frequencies of the number of components: (a) four variables (—, SHT by using ave-rk⁺; — — —, AIC by using ave-rk⁺; · · · · ·, BIC by using ave-rk⁺; · · · · ·, HQ by using ave-rk⁺; —○—, AIC by using the maximum likelihood estimator; —○—, BIC by using the maximum likelihood estimator; ···○···, HQ by using the maximum likelihood estimator); (b) eight variables (—, SHT by using max-rk⁺; — — —, mean of 70 SHTs)

in partition 3; using a larger number of partitions does not necessarily improve performance because $\text{var}(\hat{P}_\Delta)$ increases with the number of partitions.

Figs 1(d), 1(e) and 1(f) show the selection frequency across different partitions when we generate samples from three-component χ^2 -mixtures, where $\pi^m = \frac{1}{3}$, $X^m \sim \chi^2(k_x^m)$, and $Y^m \sim \chi^2(k_y^m)$ with $(k_x^1, k_y^1) = (1, 1)$, $(k_x^2, k_y^2) = (3, 6)$ and $(k_x^3, k_y^3) = (6, 3)$. Figs 1(g), 1(h) and 1(i) report the selection frequency for three-component gamma mixtures with component-specific shape parameters, where $\pi^m = \frac{1}{3}$, $X^m \sim \text{gamma}(k_1^m, 1)$ and $Y^m \sim \text{gamma}(k_2^m, 1)$ with $(k_1^1, k_2^1) = (1, 1)$, $(k_1^2, k_2^2) = (1.5, 3)$ and $(k_1^3, k_2^3) = (3, 1.5)$. In both the χ^2 - and the gamma mixtures, the relative performance across the SHT, AIC, BIC and HQ test and the relative performance of the SHT across different partitions are qualitatively similar to the case of the normal mixture that was discussed above.

Next, we consider a four-variable, three-component normal mixture, where $W = (W_1, \dots, W_4)'$ follows $\sum_{m=1}^M \pi^m N_4(\mu^m, I_4)$ with $\mu^1 = (0, 0, 0, 0)'$, $\mu^2 = (1.0, 2.0, 0.5, 1.0)'$, $\mu^3 = (2.0, 1.0, 1.0, 0.5)'$ and $\pi^1 = \pi^2 = \pi^3 = \frac{1}{3}$. Following the approach in Section 3.4, we consider three groupings: $\{X^1, Y^1\} = \{(W_1, W_2), (W_3, W_4)\}$, $\{X^2, Y^2\} = \{(W_1, W_3), (W_2, W_4)\}$ and $\{X^3, Y^3\} = \{(W_1, W_4), (W_2, W_3)\}$. We then estimate the probability matrix P_Δ^α for each $\alpha \in \{1, 2, 3\}$ and construct the ave-rk⁺-statistic (11) by setting c equal to 0.01 times the largest eigenvalue of $\Omega_r(A_0)$. The support of W_i is partitioned into two equiprobable subsets based on its empirical median, so that the dimension of P_Δ^α is 4×4 . As an alternative method, we also consider the maximum likelihood estimator based parametric model selection procedure with the AIC, BIC and HQ criterion, where each component distribution is correctly specified as a four-dimensional normal distribution with unknown means and an unknown diagonal covariance matrix. Fig. 2(a) reports the result. The maximum likelihood estimator based AIC substantially overestimates the number of components. Although the maximum likelihood estimation based HQ criterion outperforms

the ave-rk⁺-based SHT, their performances are comparable when $N \geq 2000$. This is encouraging given that our ave-rk⁺-based methods do not use parametric restrictions of the normal mixture model.

We also consider an eight-variable, three-component normal mixture, where $W = (W_1, \dots, W_8)'$ follows $\sum_{m=1}^3 \pi^m N_8(\mu^m, I_8)$ with $\mu^1 = (0, 0, 0, 0, 0, 0, 0, 0)'$, $\mu^2 = (1.0, 2.0, 0.5, 1.0, 0.75, 1.25, 0.25, 0.5)'$, $\mu^3 = (2.0, 1.0, 1.0, 0.5, 1.25, 0.75, 0.5, 0.25)'$ and $\pi^1 = \pi^2 = \pi^3 = \frac{1}{3}$. We first choose four variables out of eight, and then construct the ave-rk⁺-statistic by using the procedure for the four-variable model that was discussed in the previous paragraph. Because $\binom{8}{4} = 70$, we construct 70 ave-rk⁺-statistics. Finally, we compute the max-rk⁺-statistic from these 70 ave-rk⁺-statistics. Fig. 2(b) reports the selection frequency of the SHT by using the max-rk⁺-statistic and the mean selection frequencies by the SHT across 70 different ave-rk⁺-statistics. The max-rk⁺-statistic performs substantially better than individual ave-rk⁺-statistics, suggesting that combining information from different ave-rk⁺-statistics improves the performance of our procedures.

5. Examples

5.1. Intergenerational occupational mobility in Great Britain

We estimate the number of latent classes in the table of intergenerational mobility from father's occupation to subject's occupation in Great Britain, which was originally studied by Clogg (1981) using latent class models. Clogg estimated the two-class and three-class models using these data by imposing *a priori* restrictions on a set of parameters. Panel (a) of Table 1 presents the result of the SHT procedure applied to the 5×5 table of social mobility in Great Britain taken from Table 1.B of Clogg (1981); the null hypothesis that the number of latent classes is no more than 4 is rejected at any level of significance. The AIC, BIC and HQ procedures also indicate that the number of latent classes is at least 5 (not reported in Table 1). As reported in panel (b) of Table 1, we further examine the number of latent classes in the 8×8 table by using Table 1.C of Clogg (1981) starting from the null hypothesis of no more than five classes. The SHT suggests that these intergenerational occupational mobility data could be generated from seven latent classes whereas the BIC, AIC and HQ test suggest five, eight and six latent classes respectively. Overall, the results of our procedures suggest that there are more than five latent classes, rejecting the two- and three-class models that were studied by Clogg.

5.2. Types of trades started by different ethnic groups

The second example analyses the difference across ethnic groups in the types of trades that they start in two large cities in the Netherlands, Amsterdam and Rotterdam, which were studied by van der Heijden *et al.* (2002). There are six types of trades and five ethnic groups for each

Table 1. Intergenerational social mobility in Great Britain†

	(a) Results for 5×5 table				(b) Results for 8×8 table			
Null hypothesis (H_0)	$M=1$	$M=2$	$M=3$	$M=4$	$M=5$	$M=6$	$M=7$	
rk-statistic	557.08	144.64	48.18	15.71	35.59	12.33	2.27	
Degrees of freedom	16	9	4	1	9	4	1	
p-value	0.000	0.000	0.000	0.000	0.000	0.015	0.132	

†The data are from Tables 1.B and 1.C of Clogg (1981).

Table 2. Type of trade and ethnic group data, Amsterdam and Rotterdam†

Values of rk-statistics and the degrees of freedom for the following cities:								
	Amsterdam				Rotterdam			
Null hypothesis (H_0)	$M = 1$	$M = 2$	$M = 3$	$M = 4$	$M = 1$	$M = 2$	$M = 3$	$M = 4$
rk-statistic	318.09	57.87	13.48	0.23	190.23	60.82	9.20	1.88
Degrees of freedom	20	12	6	2	20	12	6	2
p -value	0.000	0.000	0.036	0.891	0.000	0.000	0.163	0.391

†The data are from Table 2a of van der Heijden *et al.* (2002).

of the two cities. The members of some ethnic groups are more likely to start certain types of trades because of such factors as the number of clients in the same ethnic group or their level of human capital, including knowledge of the Dutch language. From this viewpoint, each latent class could be reflecting a specific type of network and human capital. On the basis of likelihood ratio statistics, van der Heijden *et al.* (2002) concluded that the number of latent classes $M = 3$ ‘seems adequate’ for both Amsterdam and Rotterdam. We apply our procedures to examine whether the number of latent classes is at least 3 or not. Table 2 shows the rk-statistics and the corresponding p -values from the SHT procedure. For Amsterdam, the SHT suggests three or four latent classes, whereas the AIC, BIC and HQ test suggest four, two and three latent classes respectively. For Rotterdam, all our procedures suggest three latent classes.

5.3. Response patterns in five-item subsets of the Law School admission test

In our third example, we analyse the response patterns in two different five-item subsets of the Law School admission test, denoted by LSAT-6 and LSAT-7, which were originally studied by Mislevy (1984). We apply the max-rk⁺-statistic to these data. The response to five items is represented by $\{W_1, W_2, W_3, W_4, W_5\}$ where $W_i \in \{0, 1\}$. We first choose four items out of five and then construct the ave-rk⁺-statistic from the estimates of P_{Δ}^{α} s for three different groupings $\alpha = 1, 2, 3$, where we estimate the covariance matrix $\Omega_{r,c}(\mathcal{A}_0)$ by using the asymptotic formula. Because there are $\binom{5}{4} = 5$ different ways of choosing four items out of five, we construct the max-rk⁺-statistic from the five ave-rk⁺-statistics. The SHT based on the max-rk⁺-statistic suggests that $M \geq 2$ in LSAT-6 at $\alpha = 0.1, 0.05, 0.01$ and that $M \geq 3$ at $\alpha = 0.1$ and $\alpha = 0.05$ and $M \geq 2$ at $\alpha = 0.01$ in LSAT-7.

5.4. Example 3 of Hettmansperger and Thomas (2000)

We also apply our procedure to the data that consist of 83 college-age women each with eight replications of Witkin’s rod-and-frame task. The response variable, which is measured as the rod’s error deviation in degrees from the vertical, is continuously distributed. Hettmansperger and Thomas (2000) applied various tests of the number of components to the data transformed to a binomial mixture: Lindsay’s (1995) gradient function method suggests $M = 4$, the Hellinger and the Pearson penalized distances suggest $M = 2$ and the bootstrapped likelihood ratio test suggests $M = 3$. Following Hettmansperger and Thomas, we use the known cut-off point of 5° to define the eight response variables $\{W_1, \dots, W_8\}$ with $W_j = \mathbf{1}(|X_j| \leq 5^\circ)$, where X_j is the j th error. We then calculate the max-rk⁺-statistic from the 70 ave-rk⁺-statistics with $c = 0.01$, each of which is constructed from the four chosen variables out of the eight response variables

by using the covariance matrix $\Omega_r(\mathcal{A}_0)$ estimated by the asymptotic formula. The SHT based on the max-rk⁺-statistic suggests that $M \geq 3$ at $\alpha = 0.1, 0.05, 0.01$, which is consistent with Hettmansperger and Thomas's result from the bootstrapped likelihood ratio test.

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References

- Allman, E. S., Matias, C. and Rhodes, J. A. (2009) Identifiability of parameters in latent structure models with many observed variables. *Ann. Statist.*, **37**, 3099–3132.
- Andrews, D. W. K. (1987) Asymptotic results for generalized Wald tests. *Econometr. Theor.*, **3**, 348–358.
- Benaglia, T., Chauveau, D. and Hunter, D. R. (2009) An EM-like algorithm for semi- and non-parametric estimation in multivariate mixtures. *J. Computat. Graph. Statist.*, **18**, 505–526.
- Bhattacharya, A. and Dunson, D. B. (2012) Simplex factor models for multivariate unordered categorical data. *J. Am. Statist. Ass.*, **101**, 362–377.
- Borkowf, C. B. (2000) On multidimensional contingency tables with categories defined by the empirical quantiles of the marginal data. *J. Statist. Plann. Inf.*, **91**, 33–51.
- Chen, J. and Kalbfleisch, J. D. (1996) Penalized minimum-distance estimates in finite mixture models. *Can. J. Statist.*, **24**, 167–175.
- Clogg, C. C. (1981) Latent structure models of mobility. *Am. J. Sociol.*, **86**, 836–868.
- Clogg, C. C. (1995) Latent class models. In *Handbook of Statistical Modeling for the Social and Behavioral Sciences* (eds G. Arminger, C. C. Clogg and M. E. Sobel), pp. 311–359. New York: Plenum.
- Cohen, J. E. and Rothblum, U. G. (1993) Nonnegative ranks, decompositions, and factorizations of nonnegative matrices. *Lin. Alg. Applic.*, **190**, 149–168.
- Cruz-Medina, I. R., Hettmansperger, T. P. and Thomas, H. (2004) Semiparametric mixture models and repeated measures: the multinomial cut point model. *Appl. Statist.*, **53**, 463–474.
- Dacunha-Castelle, D. and Gassiat, E. (1999) Testing the order of a model using locally conic parametrization: population mixtures and stationary ARMA processes. *Ann. Statist.*, **27**, 1178–1209.
- Dong, B., Lin, M. M. and Chu, M. T. (2009) Nonnegative rank factorization via rank reduction. *Preprint*. North Carolina State University, Raleigh.
- Dunson, D. B. and Xing, C. (2009) Nonparametric Bayes modeling of multivariate categorical data. *J. Am. Statist. Ass.*, **104**, 1042–1051.
- Elmore, R. T., Hettmansperger, T. P. and Thomas, H. (2004) Estimating component cumulative distribution functions in finite mixture models. *Commun. Statist. Theor. Meth.*, **33**, 2075–2086.
- Everitt, B. S. and Hand, D. J. (1981) *Finite Mixture Distributions*. New York: Wiley.
- Goodman, L. A. (1974) The analysis of systems of qualitative variables when some of the variables are unobservable: part I, a modified latent structure approach. *Am. J. Sociol.*, **79**, 1179–1259.
- Hagenaars, J. A. and McCutcheon, A. L. (2002) *Applied Latent Class Analysis*. Cambridge: Cambridge University Press.
- Hall, P., Neeman, A., Pakyari, R. and Elmore, R. (2005) Nonparametric inference in multivariate mixtures. *Biometrika*, **92**, 667–678.
- Hall, P. and Zhou, X.-H. (2003) Nonparametric estimation of component distributions in a multivariate mixture. *Ann. Statist.*, **31**, 201–224.
- van der Heijden, P. G. M., van der Ark, L. A. and Mooijaart, A. (2002) Some examples of latent budget analysis and its extensions. In *Applied Latent Class Analysis* (eds J. A. Hagenaars and A. L. McCutcheon), pp. 107–136. Cambridge: Cambridge University Press.
- Henna, J. (1985) On estimating of the number of constituents of a finite mixture of continuous distributions. *Ann. Inst. Statist. Math.*, **37**, 235–240.
- Hettmansperger, T. P. and Thomas, H. (2000) Almost nonparametric inference for repeated measures in mixture models. *J. R. Statist. Soc. B*, **62**, 811–825.
- James, L. F., Priebe, C. E. and Marchette, D. J. (2001) Consistent estimation of mixture complexity. *Ann. Statist.*, **29**, 1281–1296.

- Kasahara, H. and Shimotsu, K. (2009) Nonparametric identification of finite mixture models of dynamic discrete choices. *Econometrica*, **77**, 135–175.
- Keribin, C. (2000) Consistent estimation of the order of mixture models. *Sankhya A*, **62**, 49–62.
- Kleibergen, F. and Paap, R. (2006) Generalized reduced rank tests using the singular value decomposition. *J. Econometr.*, **133**, 97–126.
- Lazarsfeld, P. F. and Henry, N. W. (1968) *Latent Structure Analysis*. Boston: Houghton Mifflin.
- de Leeuw, J. and van der Heijden, P. G. M. (1988) The analysis of time-budgets with a latent time-budget model. In *Data Analysis and Informatics 5* (ed. E. Diday), pp. 159–166. Amsterdam: North-Holland.
- Leroux, B. G. (1992) Consistent estimation of a mixing distribution. *Ann. Statist.*, **20**, 1350–1360.
- Levine, M., Hunter, D. R. and Chauveau, D. (2011) Maximum smoothed likelihood for multivariate mixtures. *Biometrika*, **98**, 403–416.
- Lim, L.-H. and Comon, P. (2009) Nonnegative approximations of nonnegative tensors. *J. Chemometr.*, **23**, 432–441.
- Lindsay, B. G. (1995) *Mixture Models: Theory, Geometry, and Applications*. Hayward: Institute of Mathematical Statistics.
- Lindsay, B. G. and Roeder, K. (1992) Residual diagnostics for mixture models. *J. Am. Statist. Ass.*, **87**, 785–794.
- Lütkepohl, H. and Burda, M. M. (1997) Modified Wald tests under nonregular conditions. *J. Econometr.*, **78**, 315–332.
- Magidson, J. and Vermunt, J. K. (2004) Latent class models. In *The Sage Handbook of Quantitative Methodology for the Social Sciences* (ed. D. Kaplan), pp. 175–198. Thousand Oaks: Sage.
- McLachlan, G. J. and Basford, K. E. (1988) *Mixture Models: Inference and Applications to Clustering*. New York: Dekker.
- McLachlan, G. and Peel, D. (2000) *Finite Mixture Models*. New York: Wiley.
- Mislevy, R. J. (1984) Estimating latent distribution. *Psychometrika*, **49**, 359–381.
- Robin, J.-M. and Smith, R. (2000) Tests of rank. *Econometr. Theor.*, **16**, 151–175.
- Roeder, K. (1994) A graphical technique for detecting the number of components in a mixture of normals. *J. Am. Statist. Ass.*, **89**, 487–495.
- Schork, N. J., Weder, A. B. and Schork, A. (1990) On the asymmetry of biological frequency distributions. *Genet. Epidemiol.*, **7**, 427–446.
- Skrondal, A. and Rabe-Hesketh, S. (2004) *Generalized Latent Variable Modeling: Multilevel, Longitudinal, and Structural Equation Models*. Boca Raton: Chapman and Hall–CRC.
- Titterton, D. M., Smith, A. F. M. and Makov, U. E. (1985) *Statistical Analysis of Finite Mixture Distributions*. New York: Wiley.
- Vavasis, S. (2009) On the complexity of nonnegative matrix factorization. *SIAM J. Optimizn*, **20**, 1364–1377.
- Windham, M. P. and Cutler, A. (1992) Information ratios for validating mixture analysis. *J. Am. Statist. Ass.*, **87**, 1188–1192.
- Woo, M.-J. and Sriram, T. N. (2006) Robust estimation of mixture complexity. *J. Am. Statist. Ass.*, **101**, 1475–1486.
- Zhou, X. H., Castelluccio, P. and Zhou, C. (2005) Nonparametric estimation of ROC curves in the absence of a gold standard. *Biometrics*, **61**, 600–609.

Supporting information

Additional 'supporting information' may be found in the on-line version of this article:

'Supplementary appendix to "Nonparametric identification and estimation of the number of components in multivariate mixtures"'.¹